Chapter 18, Problem 32P

Problem

Determine the functions corresponding to the following Fourier transforms:

(a) $F_1(\omega) = \dfrac{e^{j\omega}}{-j\omega + 1}$

(b) $F2(\omega) = 2e|\omega|$

(c) $F_3(\omega) = \dfrac{1}{(1+\omega^2)^2}$

(d) $F_4(\omega) = \dfrac{\delta(\omega)}{1 + j2\omega}$

Step-by-step solution

Step 1 of 4

(A) $F_1(\omega) = \dfrac{e^{j\omega}}{-j\omega+1} = \dfrac{e^{j\omega}}{j\omega - 1}$

$\Rightarrow \boxed{f(t) = -e^{t}u(t+1)}$

Step 2 of 4

(B) $F_2(\omega) = 2e^{|\omega|}$

$$0r, \qquad F_2(\omega) = \begin{cases} 2e^{\omega} & \omega > o \\ 2e^{-\omega} & \omega < 0 \end{cases}$$

This equation can be writtren as

$F_2(\omega) = 2e^{\omega}.1 + 2e^{-\omega}.1$

$\boxed{f_2(t) = 2\delta(t-1) + 2\delta(-t+1)}$

Step 3 of 4

(C) $F_3(\omega) = \dfrac{1}{(1+\omega^2)^2} = \dfrac{1}{\left[(1+j\omega)(1-j\omega)\right]^2} = \dfrac{1}{(1+j\omega)^2} * \dfrac{1}{(1-j\omega)^2}$

$F_1^{'}(\omega) * F_2^{'}(\omega)$

$\Rightarrow \boxed{f_3(t) = te^{-t}\ \ u(t) * te^{t}\ \ u(-t)}$

Step 4 of 4

(D) $F_4(\omega) = \frac{\delta(\omega)}{1+2j\omega}$

$= \frac{2\pi\ \delta(\omega)}{2\pi} * \frac{1}{1+2j\omega}$

$\Rightarrow f_4(t) = \frac{1}{2\pi}.1 * e^{-2t}u(t)$

$\boxed{f_4(t) = \frac{e^{-2t}}{2\pi}u(t)}$